

PDFEdits — AWS Production Architecture

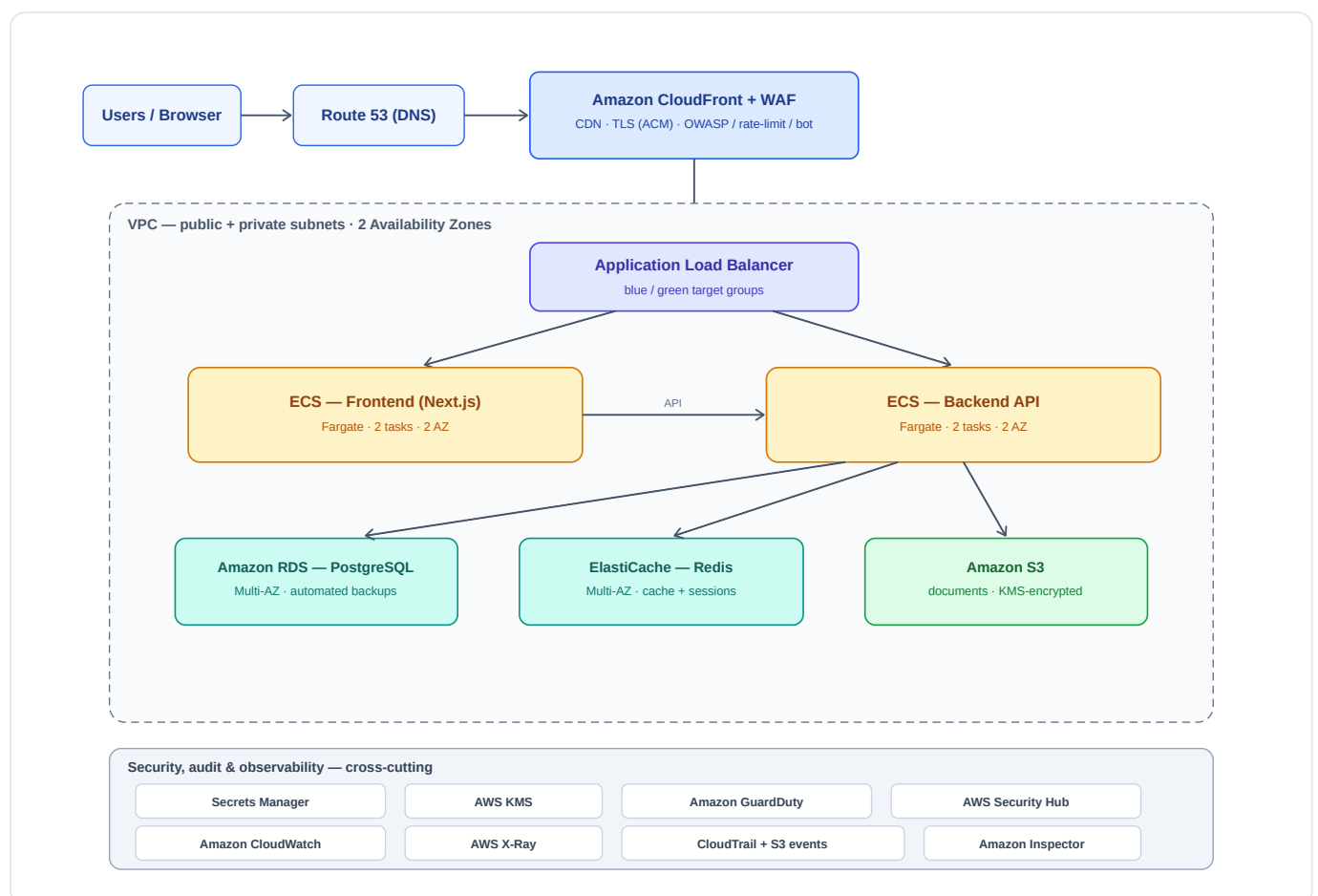
Scale target: 10,000 users/day · vertical scaling for growth · Dockerized frontend + backend

Solution Architecture Review · Confidential — for client review | Region: ap-south-1 (Mumbai) | June 2026

1 Objective & scope

A production-grade design for PDFEdits sized for **10,000 users/day**, with headroom to **scale vertically** as traffic grows. The application is two Dockerized services — a **frontend** and a **backend** (no separate PDF worker) — on ECS Fargate across two Availability Zones. It delivers the four requirements: **full logging & monitoring**, **object-level S3 audit via CloudTrail**, a **proper security posture**, and **zero-downtime deployments with automatic rollback**. The load balancer is retained because blue/green deployment and health-checked routing depend on it. As demand increases, resource sizes step up — and ultimately scale out — without changing this architecture.

2 Reference architecture



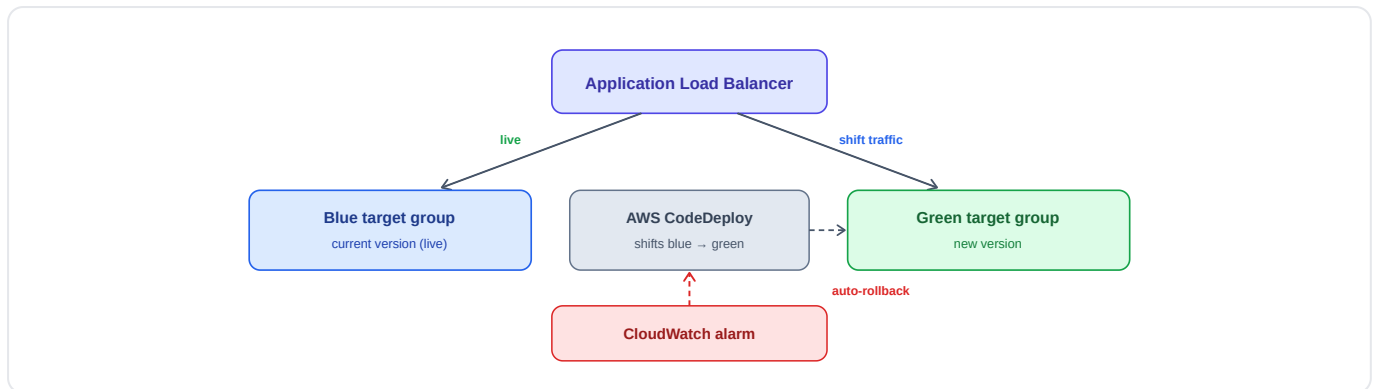
3 Component plan & monthly cost

Estimates in USD per month. Basis: region ap-south-1 (Mumbai), on-demand pricing, ~10,000 users/day (~ 6M requests/month). Current AWS list pricing, June 2026.

Service	Role	Configuration	Est. \$/mo
EDGE & DNS			
Route 53	DNS & health-checked routing.	1 zone + ~5M queries.	\$5
Amazon CloudFront	CDN, TLS, edge caching, public entry point.	~300 GB India egress + ~6M requests.	\$35
ACM	Public TLS certificate.	Free.	Free
AWS WAF	OWASP, rate-limiting and bot rules at the edge.	1 web ACL, ~12 rules, ~6M requests.	\$25
NETWORK			
Amazon VPC	Public + private subnets, 2 AZ.	Security groups + NACLs (VPC is free).	\$0
NAT Gateway	Outbound internet for private subnets.	2 AZ.	\$70
VPC Endpoints	PrivateLink to S3 / ECR / Secrets / CloudWatch.	~4 services × 2 AZ.	\$45
Data transfer	Cross-AZ traffic (Multi-AZ HA).	Inter-AZ at \$0.01/GB.	\$15
COMPUTE			
Amazon ECS (Fargate)	Frontend + backend containers, 2 tasks each, 2 AZ.	~3 vCPU / 6 GB total (scale vertically).	\$120
Application Load Balancer	Path routing + blue/green target groups.	24/7 + ~3 LCU average.	\$40
DATA & STATE			
Amazon RDS — PostgreSQL	Primary datastore; Multi-AZ with automated backups.	db.t4g.medium Multi-AZ + 50 GB gp3.	\$115
Amazon ElastiCache — Redis	Cache, sessions, and rate-limit counters.	cache.t4g.small × 2 (primary + replica).	\$60
Amazon S3	Document storage via pre-signed URLs.	~200 GB + requests; versioned, SSE-KMS.	\$25
SECURITY & AUDIT			
AWS Secrets Manager	Secret storage with automatic rotation.	~10 secrets + API calls.	\$10
AWS KMS	Customer-managed keys (S3, RDS, Redis, secrets).	~5 CMKs + requests.	\$10
Amazon GuardDuty	Threat detection (logs, network, DNS).	Account-wide; scales with traffic.	\$40
AWS Security Hub	Posture / CIS benchmarking.	Account-wide.	\$25
Amazon Inspector	Container image CVE scanning.	Images scanned on push to ECR.	\$8
AWS CloudTrail	API audit + S3 object-level (data) events.	Mgmt trail (free) + S3 data events.	\$10
OBSERVABILITY			
Amazon CloudWatch	Logs, metrics, dashboards, alarms.	~40 GB logs + metrics + canaries.	\$45
AWS X-Ray	Distributed tracing across services.	Sampled traces.	\$10
VPC Flow Logs	Network-level traffic logging.	To CloudWatch / S3.	\$10
Amazon SNS	Alarm / incident notifications.	Low volume.	\$2
DELIVERY			
CI/CD (CodeBuild + CodeDeploy)	Build, scan, blue/green deploy.	Build minutes; CodeDeploy to ECS is free.	\$15
Amazon ECR	Private container image registry.	~20 GB of images.	\$3
Estimated monthly total — production (on-demand)			≈ \$745

4 Zero-downtime deployment & rollback

Deployments use **AWS CodeDeploy blue/green on ECS**. The new version is started on a separate "green" target group; the load balancer shifts traffic from "blue" (current) to green only after health checks pass; and any failure during the shift rolls back to blue automatically. Users never see downtime or a bad release.



- **Blue/green shift.** New tasks register to the green target group; CodeDeploy moves ALB traffic from blue to green — all-at-once, or gradually (canary / linear) for extra safety.
- **Validation gate.** Traffic shifts only after green passes ALB health checks; you can run smoke tests against green before the cut-over.
- **Automatic rollback.** A CloudWatch alarm (e.g. a 5xx or latency spike) during the deployment triggers an immediate rollback to blue — zero downtime, no manual step.
- **Promotion path.** The pipeline deploys to **staging first, then production**: GitHub push → build image → Inspector scan → push to ECR → CodeDeploy blue/green.

5 Security posture

- **Edge protection.** AWS WAF on CloudFront (OWASP top-10, rate limiting, bot control). Shield Standard is always on; Shield Advanced (managed DDoS) is an optional add-on (see §8).
- **Network isolation.** ECS tasks and the data tier live in **private subnets** across 2 AZ — only the ALB is public. Security groups and NACLs enforce least-path access; NAT Gateway handles outbound; VPC endpoints keep S3 / ECR / Secrets / CloudWatch traffic off the public internet.
- **Secrets.** AWS Secrets Manager with automatic rotation — no secrets in the image or code; encrypted with KMS.
- **Encryption.** KMS customer-managed keys for S3, RDS, ElastiCache and secrets at rest; TLS end-to-end in transit.
- **Least privilege.** Per-service IAM task roles scoped to exactly what each service needs — frontend and backend cannot assume each other's permissions.
- **Threat detection.** Amazon GuardDuty (account, network, DNS) + AWS Security Hub (posture scoring) + Amazon Inspector (image CVEs).

6 Logging, monitoring & audit

- **Metrics & alarms.** Amazon CloudWatch dashboards and alarms on CPU, memory, request count, p95 latency and 5xx rate → SNS notifications (Slack / email / PagerDuty).
- **Centralized logs.** Container logs from both services to CloudWatch Logs; ALB and CloudFront access logs to S3; VPC Flow Logs for network-level visibility.
- **Distributed tracing.** AWS X-Ray follows each request end-to-end — frontend → backend → RDS / Redis / S3 — to pinpoint latency and errors.
- **Uptime monitoring.** CloudWatch Synthetics canaries continuously exercise key user journeys and alarm on failure (these also drive deployment rollback).
- **Account & data audit.** AWS CloudTrail records every API action, and **S3 data events are enabled** so every object-level operation — GetObject, PutObject, DeleteObject — on the documents bucket is logged. This is the per-S3-event audit you asked for. Note that because it logs every object access, its cost scales with traffic.

7 Cost summary & scaling

Estimated monthly total — production, on-demand (~10K users/day)

≈ \$745 / month

- **With 1-year Savings Plans** on the Fargate, RDS and ElastiCache lines, the steady-state drops roughly 30–40% to about **\$580–650/month**.
- **A large share is a fixed HA / security floor** — NAT Gateway, VPC endpoints, ALB, the Multi-AZ database and cache, GuardDuty and Security Hub stay roughly flat regardless of traffic. Dropping to single-AZ data and fewer endpoints would cut cost further, at the expense of the HA / security posture.
- **With AWS Shield Advanced** (optional managed DDoS, \$3,000/mo flat) the total is ≈ **\$3,745/month** — only worth it if you are a genuine DDoS target.

Scaling for more traffic — vertical first

As usage grows, step up sizes with no architectural change: larger Fargate task CPU/memory, a larger RDS class (db.t4g.medium → db.m6g.large → db.r6g.large), and a larger ElastiCache node. When vertical headroom runs out, scale out — more Fargate tasks, an RDS read replica (or Aurora), and a larger CloudFront tier. The same design carried all the way to the 100K-users/day sizing.

8 Notes

- **Two environments.** Run staging and production as separate, isolated stacks (ideally separate AWS accounts under one Organization). Staging is a smaller replica — minimal ECS tasks, a single-AZ database, container Redis — at roughly \$30–60/month; the same blue/green pipeline promotes from staging to production.
- **No PDF worker.** PDF operations run inside the backend service. If server-side processing later becomes heavy, the clean step is to offload it to a queue (SQS) + a dedicated worker fleet — without changing the rest of this design.
- **Dockerization.** Build one image per service (frontend, backend), scanned by Inspector and stored in ECR; ECS Fargate runs them with no servers to manage.
- **HA is preserved at 10K.** Two tasks per service across 2 AZ, a Multi-AZ database, and a replica cache keep the zero-downtime and failover guarantees intact even at the smaller size.